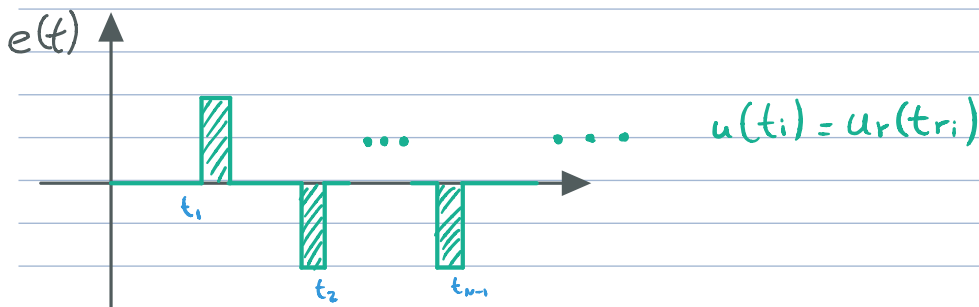
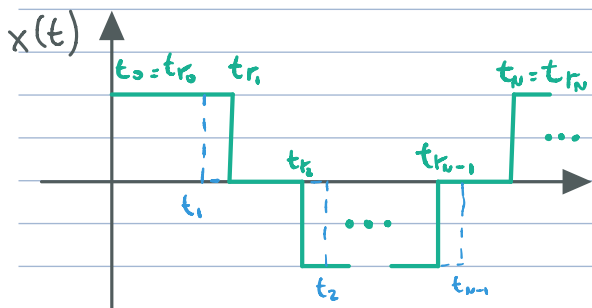


Reunião 29 jun 2021



$$e(t_N) = e^{A_c T} t_0 + \int_{t_0}^{t_N} e^{A_c(t_N - \tau)} B_c [u(\tau) - u_r(\tau)] d\tau$$

$$\int_{t_0}^{t_N} e^{A_c(t_N - \tau)} B_c [u(\tau) - u_r(\tau)] d\tau =$$

$$= \sum_{i=1}^{N-1} \int_{t_{r_i}}^{t_i} e^{A_{c_i}(t_N - \tau)} B_{c_i} [u_r(t_{r_{i-1}}) - u_r(t_{r_i})] d\tau$$

$$= - \sum_{i=1}^{N-1} \int_{t_{r_i}}^{t_i} e^{A_{c_i}(t_N - \tau)} B_{c_i} \Delta u_r(t_{r_i}) d\tau$$

Linearização

$$\int_{t_a}^{t_b} e^{-A_c \tau} d\tau \doteq e^{-A_c t_a} (t_b - t_a) \quad (\text{Kieck, 2004})$$

$$\int_{t_{r_i}}^{t_i} e^{A_{c_i}(t_N - \tau)} d\tau = e^{A_{c_i} t_N} \int_{t_{r_i}}^{t_i} e^{-A_{c_i} \tau} d\tau \doteq e^{A_{c_i} t_N} \cdot e^{-A_{c_i} t_{r_i}} (t_i - t_{r_i}) =$$

$$= e^{A_0 t_n} \cdot e^{-A_0 t_i} (t_i - t_{i-1}) = e^{A_0 (t_n - t_i)} (t_i - t_{i-1})$$

$$\int_{t_{i-1}}^{t_i} e^{A_0 (t_n - \tau)} d\tau \doteq \Delta_i \delta t_i$$

$$= - \sum_{i=1}^{N-1} \int_{t_{i-1}}^{t_i} e^{A_0 (t_n - \tau)} B_0 \Delta u_r(t_i) d\tau = - \sum_{i=1}^{N-1} \left[\int_{t_{i-1}}^{t_i} e^{A_0 (t_n - \tau)} d\tau \right] B_0 \Delta u_r(t_i)$$

$$= - \sum_{i=1}^{N-1} \Delta_i B_0 \Delta u_r(t_i) \delta t_i$$

$$e(t_n) = e^{A_c T} t_0 + \int_{t_0}^{t_n} e^{A_c (t_n - \tau)} B_c [u(\tau) - u_r(\tau)] d\tau$$



$$e(t_n) = e^{A_c T} t_0 - \sum_{i=1}^{N-1} \Delta_i B_0 \Delta u_r(t_i) \delta t_i, \quad \Delta_i = e^{A_0 (t_n - t_i)}$$

$$A_N A_{N-1} \dots A_1 = e^{A_c \Delta t_{nN}} e^{A_c \Delta t_{nN-1}} \dots e^{A_c \Delta t_{r1}} = e^{A_c (\Delta t_{nN} + \Delta t_{nN-1} + \dots + \Delta t_{r1})} = e^{A_c T}$$

(Mathews, 2018, eq. 3.54, pg 56)

Posso dizer isso?

$$e^{A_c T} = A_N A_{N-1} \dots A_1 = e^{A_{0N} \Delta t_{nN}} e^{A_{0N-1} \Delta t_{nN-1}} \dots e^{A_{01} \Delta t_{r1}}$$